CORAL: Towards Autonomous Multi-Agent Evolution for Open-Ended Discovery

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Research Questions

- Can stronger performance emerge when a greater part of the evolutionary algorithm is delegated to autonomous agents?
- Can multiple autonomous agents scale more effectively through horizontal parallelism by exploring in parallel and exchanging discoveries?
- How do knowledge accumulation and multi-agent co-evolution contribute to improved open-ended discovery performance?

Core Idea

Current LLM-based evolutionary search relies on fixed heuristics that determine which solutions to build upon, when to test, and what knowledge to preserve. CORAL replaces this rigid control with autonomous agents that make these decisions themselves through shared persistent memory containing attempts, notes, and skills. Multiple agents explore different directions asynchronously while sharing discoveries, creating emergent behaviors like technique diffusion and cross-referencing that aren't hardcoded. The key insight is that agent autonomy over the evolutionary process leads to much higher improvement rates because agents can adaptively choose what to explore based on their analysis of prior work, rather than following predetermined rules.

Key Formula

Evaluator Function

$$E(x, y) := (s, f)$$

E is the task evaluator that takes task description x and candidate solution y, returning score s and auxiliary feedback f. This defines the evaluation interface that agents use to assess solution quality in open-ended discovery tasks.

Key Figure

Preprint. Under review.

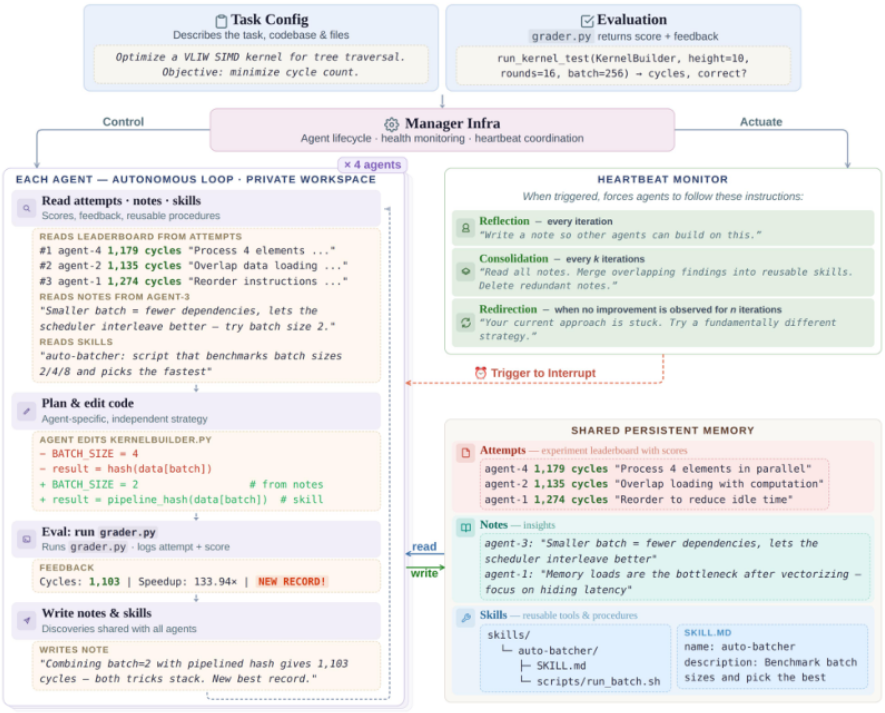

Figure 2: Overview of the CORAL framework. Autonomous agents operate in isolated worktrees, iteratively propose and evaluate candidate solutions, and accumulate shared persistent memory (attempts, notes, skills) through a hub. Heartbeat-driven periodic reflections help agents consolidate discoveries and reorient search over long horizons.

Figure 2: Shows CORAL's architecture with autonomous agents in isolated worktrees sharing persistent memory (attempts, notes, skills) through a hub, illustrating the core multi-agent coordination mechanism.

Limitations

- Relies on frontier foundation models that can handle complex coding workflows, making local deployment difficult
- Multi-agent evolution lacks bootstrapped heterogeneity as all agents are initialized identically
- Assumes availability of well-specified evaluators, which may be incomplete or ambiguous for many open-ended problems
- Current heartbeat mechanisms are predefined rather than learned or adaptive to task characteristics